

BPhO ROUND 0 · QUESTION PAPER

BPhO Round 0 — 2025 sample sheet

12 questions · 29 minutes · single answer, 5 options · **no calculator** · no negative marking

The 12 questions BPhO published as a sample. Reproduced from the official material for personal study.

Before you start

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

Q1 · Nuclide notation: which of these two nuclei are alike? K

Two nuclei, $^{3x}_{2x}\text{Q}$ and $^{4x}_{3x}\text{R}$ (x is an integer), have the same:

- | | |
|--------------------------|-------------------------|
| A specific charge | B neutron number |
| C proton number | D mass |
| E radius | |
-

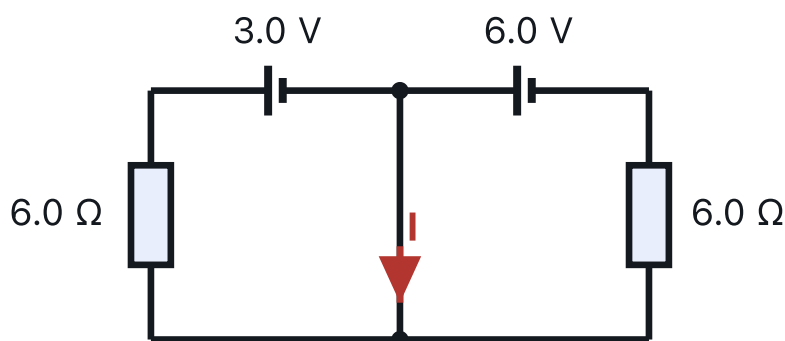
Q2 · Counting half-wavelengths on a string with five antinodes F

A string has two fixed ends, separated by a distance of 10 m. A stationary wave on the string has five antinodes. What is the wavelength?

- | | |
|--------------|--------------|
| A 2 m | B 4 m |
| C 5 m | D 6 m |
| E 8 m | |
-

Q3 · Two cells and two resistors sharing a zero-resistance link H

Calculate the current labelled **I** in the circuit below.



A 0.50 A

B 0.75 A

C 1.0 A

D 1.5 A

E 3.0 A

Q4 · Fringe spacing when the whole double-slit apparatus goes under water G

A double-slit diffraction experiment, using monochromatic light, is carried out in air and the fringe spacing w on a distant screen is measured. An identical experiment is set up under still water, which has refractive index n .

What is the fringe spacing that is measured on the underwater screen?

A w / n^2

B w / n

C w

D nw

E n^2w

Q5 · Power in one resistor: same two resistors in series, then in parallel H

A cell of internal resistance r is connected across two identical resistors, each of resistance $4r$, in series. The power dissipated in one of the resistors is P_1 . The cell is removed and connected across the same two resistors, but this time they are in parallel. The power dissipated in one of the resistors is now P_2 .

Find P_1 / P_2 .

A 1

B $4 / 9$

C $9 / 16$

D $3 / 4$

E $16 / 81$

Q6 · Which combination of units is a farad? **I**

Which one of the following is a unit of capacitance?

[Hint: capacitance is the charge stored per unit potential difference.]

A V m^{-1}

B $\text{C}^2 \text{N}^{-1}$

C $\text{V}^2 \text{s J}^{-1}$

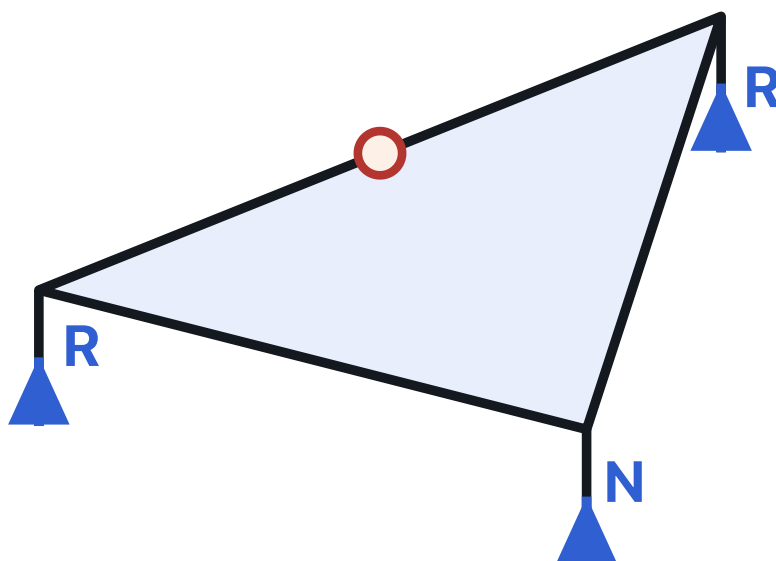
D $\text{s } \Omega^{-1}$

E $\text{A}^2 \text{s}^3 \text{J}^{-1}$

Q7 · Reactions at three rods when a weight is added halfway along an edge **C**

A uniform rigid metal sheet in the shape of an equilateral triangle has weight **W** and is supported on horizontal ground by three light inextensible rods of equal length at each of its corners. A weight **W** (same weight as the sheet) is placed on top of the sheet halfway between two corners, as shown below.

Find the ratio of reaction forces **R : N** acting on the rods from the ground.



A 5 : 1

B 4 : 1

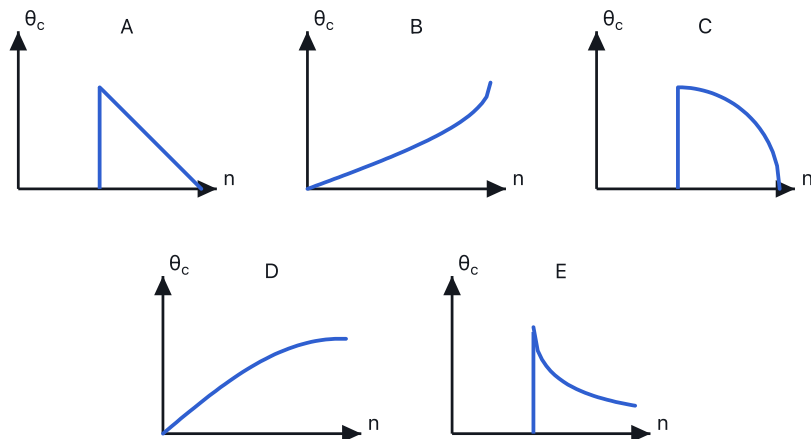
C 3 : 1

D 5 : 2

E 2 : 1

Q8 · Which sketch shows how the critical angle varies with refractive index? G

Consider the boundary between a material with refractive index n and air. Which of the following graphs best describes how the critical angle θ_c changes with n ?

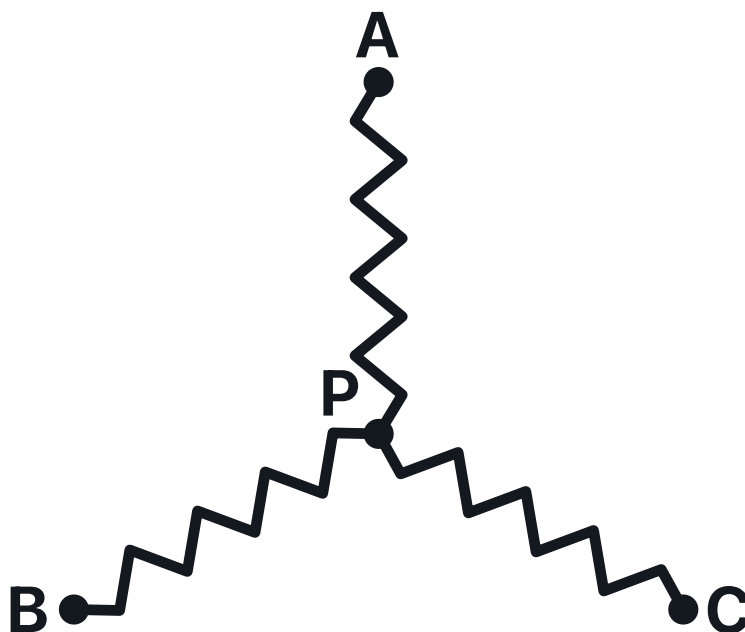


- A** A — falls in a straight line to the axis
- B** B — rises, steepening as it goes
- C** C — runs almost flat, then plunges to the axis
- D** D — rises, then levels off
- E** E — falls steeply at first, then flattens without reaching the axis
-

Q9 · Energy stored when three springs are displaced sideways by a small amount **C**

Three ideal springs of spring constant **k** and natural length **a** are arranged equidistant from a point **P**, as shown below. The points **A**, **B** and **C** are fixed in the plane of the paper, and are initially each a distance **a** away from **P**. The point **P** is then moved a small distance **d** perpendicular to the plane of the paper.

Which of these options gives the best approximation for the total energy stored in the system?



A $kd^4 / (2a^2)$

B $3kd^4 / a^2$

C $3kd^4 / (2a^2)$

D $3kd^4 / (4a^2)$

E $3kd^4 / (8a^2)$

Q10 · How the ring frequency of a rod depends on its length **A**

The frequency **f** produced by a uniform metal rod when hit by a hammer is dependent only upon its length **L**, density **ρ** and Young's modulus **E**. Find the proportionality relation between **f** and **L**.

A $f \propto 1 / L^2$

B $f \propto 1 / L$

C $f \propto 1 / \sqrt{L}$

D $f \propto \sqrt{L}$

E $f \propto L$

Q11 · Why the ground pushes less hard on a rock at the equator **D**

A small rock sitting at the Earth's North pole experiences a normal reaction force N_1 from the ground. The same rock at the equator experiences a normal reaction force N_2 . The Earth's radius is $6 \times 10^6 \text{ m}$.

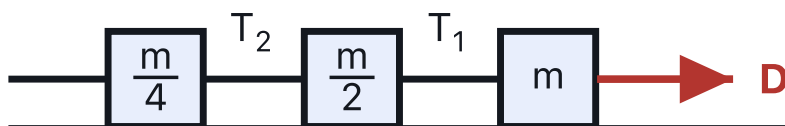
Which of these is the best estimate for $(N_1 - N_2) / N_1$?

- A** 3% **B** 0.3%
C 0.03% **D** 0.003%
E 0.0003%

Q12 · Tension in a chain whose blocks halve in mass going back **C**

A long chain of blocks, connected by light inextensible strings, travels without friction in a straight line on horizontal ground. The front block is being pulled with a force D , causing the entire chain to accelerate at a rate a . The block at the front has mass m , and each other block has half the mass of the block directly in front of it. The tensions in the strings are labelled $T_1, T_2, \dots$ from the front, as shown in the diagram.

Show that $T_n = D - f(n)ma$, and select the correct form of the function $f(n)$.



- A** $2 - n$ **B** 2^{1-n}
C $2 - 2^{-n}$ **D** $2 - 2^{1-n}$
E $1 - 2^{-n}$